

WORKED EXAMPLE

One loss, five learning rates

The starting point stays fixed so we can isolate the effect of step size.

$$E(w) = 2(w - 1)^2 = \frac{4}{2}(w - 1)^2, \quad w_0 = 3$$

$$g(w) = \frac{dE}{dw} = 4(w - 1), \quad w^* = 1$$

$$w_{k+1} = w_k - \eta g(w_k)$$

Eta is the learning rate: it multiplies the gradient to determine the parameter displacement. Iteration $k=0$ is initialization; $k=1$ means one completed update.

The minimum is at $w^* = 1$, where loss and gradient are zero. The curvature, or second derivative, is $a = 4$. This quadratic is the N4.2 study example.

WORKED EXAMPLE

Derive the signed parameter-error factor

Error here is the signed difference from the optimal parameter.

$$e_k = w_k - 1$$

$$w_{k+1} - 1 = w_k - 1 - 4\eta(w_k - 1)$$

$$e_{k+1} = (1 - 4\eta)e_k = re_k, \quad r = 1 - 4\eta$$

A positive r keeps the error on the same side. A negative r switches sides of the minimum. Its magnitude tells us whether the distance shrinks or grows.

$$E_k = 2e_k^2 \implies \frac{E_{k+1}}{E_k} = r^2$$

The loss ratio is defined when the old loss is nonzero. Absolute parameter-error ratio is $|r|$; do not confuse it with the squared ratio for loss.

WORKED EXAMPLE

Learning rate 0.1: same-side convergence

Evaluate each new gradient at the latest parameter, not at the original point.

$$g_0 = 4(3 - 1) = 8$$

$$w_1 = 3 - 0.1(8) = 2.2, \quad E_1 = 2(2.2 - 1)^2 = 2.88$$

$$g_1 = 4(2.2 - 1) = 4.8$$

$$w_2 = 2.2 - 0.1(4.8) = 1.72, \quad E_2 = 2(1.72 - 1)^2 = 1.0368$$

$$r = 1 - 4(0.1) = 0.6, \quad |r| = 0.6, \quad r^2 = 0.36$$

$$E_0 = 2(3 - 1)^2 = 8$$

Compare the distance from 1 after each update, as well as the loss.

WORKED EXAMPLE

Learning rate 0.25: minimum in one step

Evaluate each new gradient at the latest parameter, not at the original point.

$$g_0 = 4(3 - 1) = 8$$

$$w_1 = 3 - 0.25(8) = 1, \quad E_1 = 2(1 - 1)^2 = 0$$

$$g_1 = 4(1 - 1) = 0$$

$$w_2 = 1 - 0.25(0) = 1, \quad E_2 = 2(1 - 1)^2 = 0$$

$$r = 1 - 4(0.25) = 0, \quad |r| = 0, \quad r^2 = 0$$

$$E_0 = 2(3 - 1)^2 = 8$$

Compare the distance from 1 after each update, as well as the loss.

WORKED EXAMPLE

Learning rate 0.4: alternating convergence

Evaluate each new gradient at the latest parameter, not at the original point.

$$g_0 = 4(3 - 1) = 8$$

$$w_1 = 3 - 0.4(8) = -0.2, \quad E_1 = 2(-0.2 - 1)^2 = 2.88$$

$$g_1 = 4(-0.2 - 1) = -4.8$$

$$w_2 = -0.2 - 0.4(-4.8) = 1.72, \quad E_2 = 2(1.72 - 1)^2 = 1.0368$$

$$r = 1 - 4(0.4) = -0.6, \quad |r| = 0.6, \quad r^2 = 0.36$$

$$E_0 = 2(3 - 1)^2 = 8$$

Compare the distance from 1 after each update, as well as the loss.

WORKED EXAMPLE

Learning rate 0.5: nonshrinking oscillation

Evaluate each new gradient at the latest parameter, not at the original point.

$$g_0 = 4(3 - 1) = 8$$

$$w_1 = 3 - 0.5(8) = -1, \quad E_1 = 2(-1 - 1)^2 = 8$$

$$g_1 = 4(-1 - 1) = -8$$

$$w_2 = -1 - 0.5(-8) = 3, \quad E_2 = 2(3 - 1)^2 = 8$$

$$r = 1 - 4(0.5) = -1, \quad |r| = 1, \quad r^2 = 1$$

$$E_0 = 2(3 - 1)^2 = 8$$

Compare the distance from 1 after each update, as well as the loss.

WORKED EXAMPLE

Learning rate 0.6: divergence

Evaluate each new gradient at the latest parameter, not at the original point.

$$g_0 = 4(3 - 1) = 8$$

$$w_1 = 3 - 0.6(8) = -1.8, \quad E_1 = 2(-1.8 - 1)^2 = 15.68$$

$$g_1 = 4(-1.8 - 1) = -11.2$$

$$w_2 = -1.8 - 0.6(-11.2) = 4.92, \quad E_2 = 2(4.92 - 1)^2 = 30.7328$$

$$r = 1 - 4(0.6) = -1.4, \quad |r| = 1.4, \quad r^2 = 1.96$$

$$E_0 = 2(3 - 1)^2 = 8$$

Compare the distance from 1 after each update, as well as the loss.

WORKED EXAMPLE

Compare paths and costs

Different parameter paths can produce exactly the same loss sequence.

eta	r	w0, w1, w2	E0, E1, E2
0.1	0.6	3, 2.2, 1.72	8, 2.88, 1.0368
0.25	0	3, 1, 1	8, 0, 0
0.4	-0.6	3, -0.2, 1.72	8, 2.88, 1.0368
0.5	-1	3, -1, 3	8, 8, 8
0.6	-1.4	3, -1.8, 4.92	8, 15.68, 30.7328

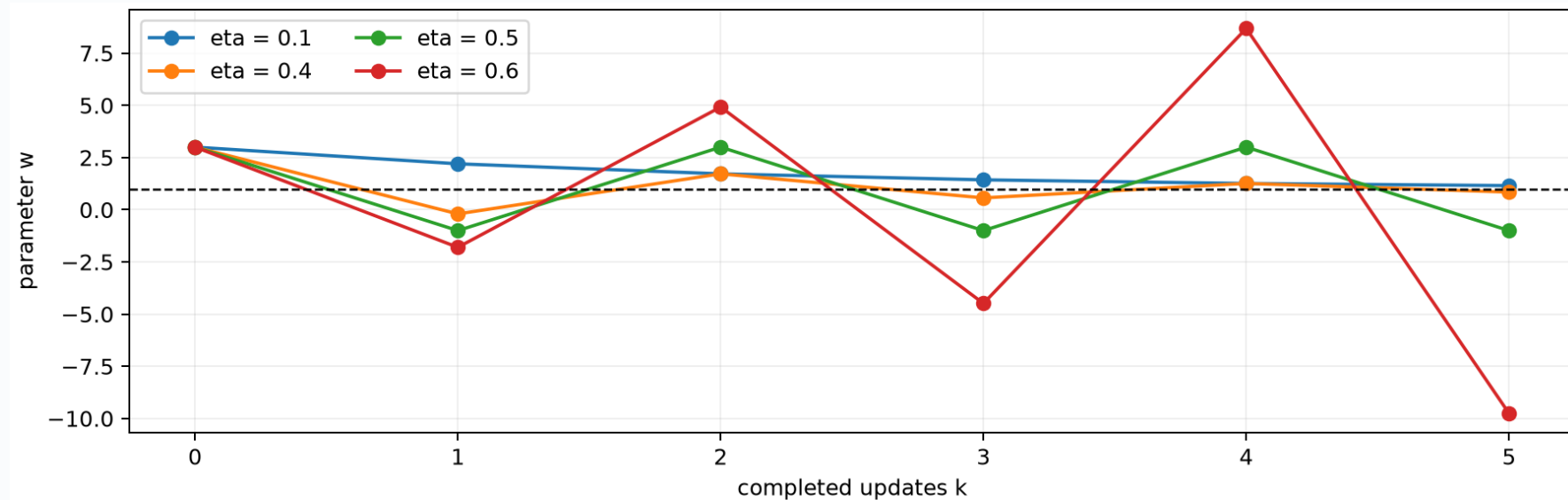
Rates 0.1 and 0.4 have the same $|r|$ and r-squared. One stays on one side; the other alternates, but their error magnitudes and losses match.

At rate 0.5, this nonoptimal starting point never converges: it keeps switching between 3 and -1. Starting at the optimum would stay there for any finite rate.

WORKED EXAMPLE

Picture the parameter paths after each update

The horizontal reference $w=1$ marks the minimum.



A crossing of the minimum does not by itself mean failure. Crossing with shrinking distance converges; crossing with constant or growing distance does not.

WORKED EXAMPLE

Derive the stable interval

We need the error magnitude to shrink after each step.

$$|r| < 1 \iff -1 < 1 - 4\eta < 1$$

$$-2 < -4\eta < 0 \iff 0 < \eta < 0.5$$

Dividing inequalities by a negative number reverses the signs. Rate zero makes no progress; the upper endpoint gives the oscillation seen earlier.

$$E = \frac{a}{2}(w - w^*)^2, \quad a > 0 \implies 0 < \eta < \frac{2}{a}$$

This statement assumes a fixed learning rate and a positive-curvature quadratic. It is not an unconditional convergence guarantee for every neural-network loss.

WORKED EXAMPLE

What linear convergence means

The name refers to a constant ratio of error magnitudes.

$$|e_{k+1}| = q|e_k|, \quad 0 < q < 1$$

$$\eta = 0.1 : \quad |e_0| = 2, \quad |e_1| = 1.2, \quad |e_2| = 0.72$$

$$q = 0.6, \quad E_1/E_0 = E_2/E_1 = 0.36$$

Each iteration removes the same fraction of the remaining distance: 40 percent here. The error shrinks geometrically, not by subtracting the same amount each time.

An alternating path can have linear convergence too: rate 0.4 also has $q=0.6$. The one-step exact solution at rate 0.25 is a special case, not an asymptotic nonzero error ratio.

WORKED EXAMPLE

Two directions can need different step sizes

A narrow bowl has one steep direction and one shallow direction.

$$E(x, y) = \frac{1}{2}(x^2 + 100y^2)$$

$$\nabla E = (x, 100y), \quad H = \text{diag}(1, 100)$$

$$x_{k+1} = (1 - \eta)x_k, \quad y_{k+1} = (1 - 100\eta)y_k$$

The diagonal Hessian has eigenvalues 1 and 100. With rate 0.01, the error factors are 0.99 and 0: the steep direction finishes immediately, while the shallow direction moves slowly.

$$0 < \eta < \min(2/1, 2/100) = 0.02$$

For a positive-definite quadratic, the largest eigenvalue sets the stable shared-rate bound $2/\lambda_{\max}$.

WORKED EXAMPLE

Work the two-direction example

Start at (1,1) and use $\eta = 0.01$ for both coordinates.

$$g_0 = (1, 100), \quad (x_1, y_1) = (1, 1) - 0.01(1, 100) = (0.99, 0)$$

$$E_0 = \frac{1+100}{2} = 50.5, \quad E_1 = \frac{0.99^2}{2} = 0.49005$$

$$g_1 = (0.99, 0), \quad (x_2, y_2) = (0.99, 0) - 0.01(0.99, 0)$$

$$= (0.9801, 0), \quad E_2 = \frac{0.9801^2}{2} = 0.480298005$$

Most of the initial loss disappears in the steep direction. After that, the shallow-direction error shrinks by only 1 percent per update. This explains why a stable rate can still be slow.

INDEPENDENT PRACTICE

Your turn: same loss speed, different paths

Use a fresh quadratic and keep initialization separate from updates.

$$F(w) = (w + 2)^2, \quad w_0 = 0$$

Calculate two updates and all three losses for rates 0.25 and 0.75. Find the minimum, curvature, signed error factors, absolute error ratios and loss ratios.

Explain why one path alternates but both losses can be identical. Derive the stable-rate interval and decide whether both chosen rates lie inside it.

WORKED ANSWER

Practice: derivatives and stability

Here the optimum is -2, so the signed error is $w+2$.

$$g(w) = 2(w + 2), \quad a = 2, \quad w^* = -2$$

$$e_{k+1} = (1 - 2\eta)e_k, \quad e_0 = 2$$

$$\eta = 0.25 : r = 0.5; \quad \eta = 0.75 : r = -0.5$$

$$|r| = 0.5, \quad r^2 = 0.25, \quad 0 < \eta < 2/2 = 1$$

Both rates are stable for this quadratic. Both halve the error magnitude and quarter the loss each step, but only the negative factor switches sides.

WORKED ANSWER

Practice: full working at rate 0.25

Evaluate the gradient again at each new parameter.

$$F_0 = (0 + 2)^2 = 4$$

$$g_0 = 2(0 + 2) = 4, \quad w_1 = 0 - 0.25(4) = -1$$

$$F_1 = (-1 + 2)^2 = 1$$

$$g_1 = 2(-1 + 2) = 2, \quad w_2 = -1 - 0.25(2) = -1.5$$

$$F_2 = (-1.5 + 2)^2 = 0.25$$

Distances from -2 are 2, 1 and 0.5. Squaring them gives losses 4, 1 and 0.25 in either path.

WORKED ANSWER

Practice: full working at rate 0.75

Evaluate the gradient again at each new parameter.

$$F_0 = (0 + 2)^2 = 4$$

$$g_0 = 2(0 + 2) = 4, \quad w_1 = 0 - 0.75(4) = -3$$

$$F_1 = (-3 + 2)^2 = 1$$

$$g_1 = 2(-3 + 2) = -2, \quad w_2 = -3 - 0.75(-2) = -1.5$$

$$F_2 = (-1.5 + 2)^2 = 0.25$$

Distances from -2 are 2, 1 and 0.5. Squaring them gives losses 4, 1 and 0.25 in either path.